# Physics Bowl Study Guide

Concepts, Methods, and Worked Examples

An independent guide for Division I and II preparation

Ray Zhou

# A Letter to the Reader

Dear Reader,

Thank you for opening this guide. It aspires to be a quiet companion on your learning journey. Physics may seem like a towering mountain from afar, but up close, it resembles a garden.

If you get stuck, sketch another diagram; study a simpler boundary case first; change only one variable at a time. Pause, ask why, and then lift your gaze to take in the scenery.

May this guide reward your Curiosity, Creativity, and Caution. And may the world hold just a little more wonder. Wishing you smooth learning and steady progress.

Ray Zhou

# How to Use These Books

1. Choose a topic in the contents. Start with the concept, check the conditions for each formula, and draw a diagram.
2. Work through the methods and pitfalls; cover the solutions and try the examples independently.
3. Follow a related-problem link to the companion problem set. Try the question before reading the key insight and solution.
4. Return to the linked unit when you miss a step. Keep track of the condition or assumption that caused the error.

**Core** and **Extension** are editorial reading cues: extensions include derivations or further applications. They are not the official contest divisions. Consult the AAPT archive for official materials.

## Notation and Conventions

- Use SI units and $g = 9.8\,\mathrm{m/s^2}$ unless a problem explicitly specifies another value. Angles are in radians unless degrees are shown.
- Arrows denote vectors, and $v = \|\vec{v}\|$ denotes a magnitude.
- $\mathcal{E}$ denotes electromotive force, while $\vec{E}$ denotes electric field. Circuit voltage is $V$.
- In thermodynamics $\Delta U = Q + W_{\text{on}}$, where $W_{\text{on}} > 0$ is work done on the system. Equivalently, $\Delta U = Q - W_{\text{by}}$.
- For thin lenses, the examples use real-object and real-image distances as positive. Check the chosen sign convention before substitution.
- Keep extra digits during calculations and round the final result consistently with the stated data.

# Contents

**A Letter to the Reader** i

**How to Use These Books** i

**Part I: Mechanics** 1

**1 Kinematics** 1
- 1.1 Vectors & Components **Core** . . . 1
- 1.2 Constant Acceleration (SUVAT) **Core** . . . 2
- 1.3 Projectile (Equal Heights) **Core** . . . 3
- 1.4 Uniform Circular Motion **Core** . . . 4
- 1.5 Relative Motion **Core** . . . 5
- 1.6 Calculus Forms & Graph Areas **Extension** . . . 6

**2 Newtonian Dynamics** 7
- 2.1 Newton's Laws and Free-Body Diagrams **Core** . . . 7
- 2.2 Friction and Springs **Core** . . . 8
- 2.3 Non-uniform Circular Motion **Core** . . . 9
- 2.4 Impulse and Variable Forces **Extension** . . . 10

**3 Work, Energy, Power** 10
- 3.1 Work and the Work–Energy Theorem **Core** . . . 11
- 3.2 Potential Energy and Conservation **Core** . . . 12
- 3.3 Conservative Fields **Extension** . . . 12

**4 Momentum & Collisions** 13
- 4.1 Momentum and Impulse **Core** . . . 13
- 4.2 Elastic and Inelastic Collisions **Core** . . . 14
- 4.3 Center of Mass and System Dynamics **Core** . . . 15
- 4.4 Variable Mass and Rockets **Extension** . . . 16

**5 Rotational Motion** 17
- 5.1 Angular Kinematics and Dynamics **Core** . . . 17
- 5.2 Angular Momentum and Conservation **Core** . . . 18
- 5.3 Moments of Inertia **Extension** . . . 19
- 5.4 Rolling Without Slipping **Core** . . . 20

**Part II: Oscillations, Waves, Thermodynamics & Fluids** 21

**6 Oscillations & Waves** 21
- 6.1 Simple Harmonic Motion **Core** . . . 21
- 6.2 Waves (Traveling) **Core** . . . 22
- 6.3 Standing Waves (Strings/Pipes) **Core** . . . 23
- 6.4 Doppler Effect (Fixed Medium) **Core** . . . 23
- 6.5 SHM ODE and Driven Forms **Extension** . . . 24

**7 Fluids & Thermodynamics** **25**
7.1 Hydrostatics and Buoyancy **Core** . . . 25
7.2 Continuity and Bernoulli **Core** . . . 26
7.3 Ideal Gas and First Law **Core** . . . 27
7.4 Entropy and Carnot **Core** . . . 28
7.5 Heat Engines and Efficiency **Core** . . . 29

**Part III: Electricity & Magnetism** **30**

**8 Electrostatics** **30**
8.1 Coulomb, Field and Potential **Core** . . . 30
8.2 Capacitors and Energy **Core** . . . 31
8.3 Gauss's Law **Core** . . . 32

**9 DC Circuits** **33**
9.1 Ohm's Law and Reductions **Core** . . . 33
9.2 Kirchhoff Laws (KCL/KVL) **Core** . . . 34
9.3 RC Transients (First Order) **Core** . . . 34

**10 Magnetism & Induction** **35**
10.1 Lorentz Force **Core** . . . 35
10.2 Magnetic Force on Wires **Core** . . . 36
10.3 Faraday-Lenz **Core** . . . 37
10.4 Ampere and Maxwell–Ampere **Extension** . . . 38
10.5 EM Spectrum and Maxwell (Concept) **Core** . . . 38

**Part IV: Optics & Modern Physics** **39**

**11 Optics** **39**
11.1 Reflection and Refraction **Core** . . . 39
11.2 Thin Lenses and Sign Conventions **Core** . . . 40
11.3 Interference and Diffraction **Core** . . . 41

**12 Modern Physics** **42**
12.1 Special Relativity **Core** . . . 42
12.2 Photoelectric Effect **Core** . . . 43
12.3 Nuclear Decay Basics **Core** . . . 43

# Part I: Mechanics

# 1 Kinematics

*Coverage:* Vectors vs scalars; constant-acceleration (SUVAT); projectile motion (same-level range/time/height); uniform circular motion ($a_c = v^2/r$, $T = 2\pi r/v$); relative motion; calculus forms $v = dx/dt$, $a = dv/dt$ **Extension**; areas under $v$–$t$ and $a$–$t$ graphs **Extension**

## 1.1 Vectors & Components **Core**

**Concepts and Formulas**

**Concept explanation:** Vectors have both magnitude and direction. The decomposition of a vector into orthogonal components translates geometric relations into algebraic relations such that calculations are simple component arithmetic.
**Core formulas:**

$$\begin{cases} \vec{v} = (v_x, v_y) \text{ (2D)}, \quad v = \|\vec{v}\| = \sqrt{v_x^2 + v_y^2}, \\ \text{Unit vectors: } \hat{\imath} = (1,0),\ \hat{\jmath} = (0,1),\ \vec{v} = v_x\hat{\imath} + v_y\hat{\jmath}. \end{cases}$$

**Variable definitions:** $v_x, v_y$ scalar components; $v$ speed (magnitude of velocity); $\hat{\imath}, \hat{\jmath}$ orthonormal basis.
**Prerequisites & scope:** Axes must be orthogonal for Pythagorean magnitude; extend to 3D with $\hat{k}$ and $v = \sqrt{v_x^2 + v_y^2 + v_z^2}$.

**Methods and Pitfalls**

- Align axes with directions of steepest slopes and launch angles to reduce components.
- Separate vectors before writing equations (Newton's laws, kinematics) to prevent late trigonometry.
- Mixing magnitude and component equations. Solution: write separate equations for $x$ and $y$ and add if applicable.

**Example 1.** Resolve velocity vector $\vec{v} = (30\,\text{m/s}, 40\,\text{m/s})$ into magnitude and direction.

**Solution**

$v = \sqrt{30^2 + 40^2} = 50\,\text{m/s}$. Direction $\theta = \arctan(40/30) \approx 53.1°$ above the horizontal.

**Example 2.** A force of 100 N acts at 30° above the positive horizontal axis. Find its horizontal and vertical components.

**Solution**

$F_x = 100\cos 30° = 50\sqrt{3}\,\text{N}$, $F_y = 100\sin 30° = 50\,\text{N}$.

## 1.2 Constant Acceleration (SUVAT) Core

**Concepts and Formulas**

**Concept explanation:** Position and velocity are quadratic/linear in time with constant acceleration, providing closed-form relations (SUVAT equations).
**Core formulas:**

$$\begin{cases} x = x_0 + v_0 t + \frac{1}{2}at^2, \\ v = v_0 + at, \\ v^2 = v_0^2 + 2a\,(x - x_0). \end{cases}$$

**Variable definitions:** $x$ position; $v$ velocity; $a$ constant acceleration; subscript 0 initial value; $t$ time; $\Delta x = x - x_0$.
**Prerequisites & scope:** Acceleration must be constant over the interval; otherwise use calculus forms.

**Methods and Pitfalls**

- Choose the equation that excludes the unknown you lack (e.g., use $v^2$-form when time is absent).
- Work symbolically until the end to avoid compounding rounding errors.
- Applying constant-$a$ formulas when $a = a(t)$ or $a = a(v)$. Fix: switch to calculus forms or energy methods.

**Example 3.** The velocity of a cart is given by the piecewise function $v(t) = \begin{cases} at, & 0 \le t \le T/2, \\ aT/2, & T/2 < t \le T, \end{cases}$ with constant $a > 0$. Find the total displacement in time $T$.

**Solution**

Area under $v$–$t$: first triangle area $\frac{1}{2}(a\,T/2)(T/2) = \frac{aT^2}{8}$; second rectangle area $(aT/2)(T/2) = \frac{aT^2}{4}$. Total $\Delta x = \frac{3aT^2}{8}$.

**Example 4.** A car accelerates uniformly from rest to 20 m/s in 4 s. Find the acceleration and distance traveled.

**Solution**

$a = \dfrac{v - v_0}{t} = \dfrac{20 - 0}{4} = 5\,\mathrm{m/s^2}$. Distance $\Delta x = v_0 t + \frac{1}{2}at^2 = 0 + \frac{1}{2} \cdot 5 \cdot 16 = 40\,\mathrm{m}$.

## 1.3 Projectile (Equal Heights) [Core]

**Concepts and Formulas**

**Concept explanation:** Horizontal and vertical motions independent in uniform gravity. Decomposition into components provides closed-form solutions for flight time, range, and peak height at the same launch/landing height.
**Core formulas:**

$$\begin{cases} T = \dfrac{2v_0 \sin\theta}{g}, \\ R = \dfrac{v_0^2 \sin 2\theta}{g}, \\ H = \dfrac{v_0^2 \sin^2\theta}{2g}. \end{cases}$$

**Variable definitions:** $v_0$ launch speed; $\theta$ launch angle; $g$ gravitational acceleration; $T$ flight time; $R$ range; $H$ apex height.
**Prerequisites & scope:** Launch/landing at equal heights; neglect air resistance; for unequal heights, solve quadratic in $t$.

**Methods and Pitfalls**

- Separate $x$ and $y$ equations; eliminate $t$ or use symmetry about the apex for time splits.
- For maximum range at equal heights, use $\theta = 45°$.
- Using $R$ formula when launch and landing heights differ. Fix: solve general quadratic and then compute $x(T)$.

**Example 5.** A projectile is launched at speed $v_0$ and angle $\theta$ from flat ground and lands at the same height. Express the time of flight $T$, the maximum height $H$, and the range $R$ in terms of $v_0, \theta, g$.

**Solution**

$T = \dfrac{2v_0 \sin\theta}{g}$, $H = \dfrac{v_0^2 \sin^2\theta}{2g}$, $R = \dfrac{v_0^2 \sin 2\theta}{g}$, by separating vertical and horizontal components and using constant-$g$ kinematics.

**Example 6.** Two projectiles launched with the same $v_0$ at angles $\theta$ and $90^\circ - \theta$ have equal ranges. What is the ratio of their maximum heights $H_\theta / H_{90^\circ-\theta}$?

**Solution**

$H = \dfrac{v_0^2 \sin^2\theta}{2g}$. Hence $\dfrac{H_\theta}{H_{90^\circ-\theta}} = \dfrac{\sin^2\theta}{\cos^2\theta} = \tan^2\theta$.

## 1.4 Uniform Circular Motion [Core]

**Concepts and Formulas**

**Concept description:** In uniform speed circular motion, the acceleration is centripetal (centerwards) with magnitude $v^2/r$; period and speed are related by circumference.

**Core formulas:**

$$\begin{cases} a_c = \dfrac{v^2}{r} = \dfrac{4\pi^2 r}{T^2}, \\ T = \dfrac{2\pi r}{v}. \end{cases}$$

**Variable definitions:** $r$ radius; $v$ speed; $T$ period; $a_c$ centripetal acceleration.
**Prerequisites & scope:** Speed constant; acceleration direction changes; for non-uniform circular motion add tangential component.

**Methods and Pitfalls**

- Draw radial/tangential components explicitly; set $a_t = 0$ for UCM.
- Vertical circle normals: at the top, $T + mg = mv^2/r$; at the bottom, $T - mg = mv^2/r$.
- At the top, a taut-string circular path requires $v_{\text{top}} \geq \sqrt{gr}$; equality is the limiting case $T = 0$. Below this speed the string cannot supply the required inward force.
- Treating centripetal force as an extra force. Fix: centripetal is the net radial component of existing forces.

**Example 7.** A bead moves in a circle of radius $r$ with period $T$. Compute its speed and centripetal acceleration.

**Solution**

$v = \dfrac{2\pi r}{T}$ and $a_c = \dfrac{v^2}{r} = \dfrac{4\pi^2 r}{T^2}$.

## 1.5 Relative Motion [Core]

### Concepts and Formulas

**Concept explanation:** Relative velocity cancels observer motion: the velocity of $A$ relative to $B$ is $\vec{v}_A - \vec{v}_B$.
**Core formula:**

$$\vec{v}_{A/B} = \vec{v}_A - \vec{v}_B.$$

**Variable definitions:** $\vec{v}_{A/B}$ velocity of $A$ relative to $B$; $\vec{v}_A, \vec{v}_B$ velocities in an inertial frame.
**Prerequisites & scope:** Within Galilean (non-relativistic) regime; in 2D/3D apply component-wise.

### Methods and Pitfalls

- Move to the target's rest frame to align directions and simplify timing.
- For winds/rivers, remove drift by aiming to oppose the current component.
- For boat speed $v$ relative to the water and current speed $u$, direct crossing requires $v > u$. If $v = u$, canceling the current leaves no speed across the river; if $v < u$, the current cannot be canceled.
- Superposing speeds scalarly when directions are different. Remedy: take difference of vectors component-wise.

**Example 8.** A river of width $W$ flows east at speed $u$. A boat of speed $v$ relative to water aims at angle $\alpha$ north of west to land directly across. Find $\alpha$ and the crossing time, assuming $v > u$.

### Solution

Require zero east drift: west component equals current, so $v\cos\alpha = u$ and $\alpha = \arccos(u/v)$. North component is $v\sin\alpha = \sqrt{v^2 - u^2}$, so time $t = \dfrac{W}{\sqrt{v^2 - u^2}}$.

## 1.6 Calculus Forms & Graph Areas [Extension]

### Concepts and Formulas

**Concept explanation:** With acceleration varying, kinematics is found by integrating velocity and acceleration; graph areas are summaries of change.
**Core formulas:**

$$\begin{cases} v = \dfrac{dx}{dt}, \quad a = \dfrac{dv}{dt} = \dfrac{d^2x}{dt^2}. \\ v(t) = v(t_0) + \displaystyle\int_{t_0}^{t} a(\tau)\,d\tau, \quad x(t) = x(t_0) + \int_{t_0}^{t} v(\tau)\,d\tau. \\ \Delta x = \displaystyle\int v\,dt \text{ (area under } v\text{–}t), \quad \Delta v = \int a\,dt \text{ (area under } a\text{–}t). \\ \text{Chain rule: } a = \dfrac{dv}{dt} = \dfrac{dv}{dx}\,v \;\Rightarrow\; v\,dv = a(x)\,dx. \\ \text{If } a = \text{const: } v^2 = v_0^2 + 2a\,(x - x_0). \end{cases}$$

**Variable definitions:** $x, v, a$ position/velocity/acceleration; $t$ time; integrals are definite over the time interval.
**Prerequisites & scope:** Differentiability over interval; interpret signed areas for direction-sensitive quantities.

### Methods and Pitfalls

- Read slopes of $x$–$t$ as $v$ and slopes of $v$–$t$ as $a$; use areas for accumulated change.
- When $a = a(v)$ or $a = a(x)$, separate variables via $v\,dv = a\,dx$ to avoid time explicitly.
- If motion crosses turning points where $v = 0$, integrate piecewise and track signs to avoid taking an incorrect branch.
- Confusing displacement with distance on $v$–$t$ when $v$ changes sign. Fix: integrate absolute value for distance.

**Example 9.** An object has acceleration $a(t) = a_0 + bt$ with constants $a_0, b$. If $x(0) = 0$ and $v(0) = v_0$, find $x(t)$.

### Solution

Integrate: $v(t) = v_0 + \int_0^t (a_0 + b\tau)\,d\tau = v_0 + a_0 t + \frac{1}{2}bt^2$. Then $x(t) = \int_0^t v(\tau)\,d\tau = v_0 t + \frac{1}{2}a_0 t^2 + \frac{1}{6}bt^3$.

**Example 10.** A particle obeys $\dfrac{dx}{dt} = k\sqrt{x}$ with $k > 0$ and $x(0) = 0$. Find all solutions for $t \geq 0$.

**Solution**

Separation is valid only where $x > 0$. The particle can remain at $x = 0$ until any $t_0 \geq 0$ and then move:

$$x(t) = \begin{cases} 0, & 0 \leq t \leq t_0, \\ \dfrac{k^2}{4}(t - t_0)^2, & t > t_0. \end{cases}$$

The identically zero solution also works. The initial value alone does not select a unique solution.

**Related problems:** Problem 1 (p. 1), Problem 10 (p. 9), Problem 13 (p. 11), Problem 18 (p. 14), Problem 26 (p. 20), Problem 27 (p. 21), Problem 30 (p. 24), Problem 35 (p. 27), Problem 39 (p. 30)

# 2 Newtonian Dynamics

*Coverage:* Newton's laws; common forces (weight, spring, friction, normal); free-body diagrams; non-uniform circular motion; variable force and impulse

## 2.1 Newton's Laws and Free-Body Diagrams Core

**Concepts and Formulas**

In an inertial frame, the net external force on a body of constant mass determines its acceleration.

$$\Big\{ \sum \vec{F} = m\vec{a} \text{ ; action–reaction pairs on separate bodies.}$$

Here $m$ is the body's mass, $\vec{a}$ its acceleration, and the sum contains forces acting on that body.

**Methods and Pitfalls**

- Split each body; sketch clean FBDs and project along convenient directions.
- Don't place action–reaction pairs on the same drawing; they're on separate bodies and cancel only at the system level.
- Use radial/tangential axes for curves or inclined axes on ramps; use friction direction as unknown and solve for its sign.
- For connected bodies, derive the displacement and acceleration constraints from constant string length; an ideal massless string has uniform tension.

- In non-inertial frames (elevators, speeding cars), add $-m\,\vec{a}_{\text{frame}}$ only if you change frames explicitly.

**Example 1.** A 5 kg block rests on a horizontal surface. A horizontal force $F = 20\,\text{N}$ is applied. Find its acceleration if friction is negligible.

**Solution**

$\sum F = ma \Rightarrow 20 = 5a \Rightarrow a = 4\,\text{m/s}^2$.

**Example 2.** A 2 kg block hangs from a string. Find the tension in the string when the block is at rest.

**Solution**

$\sum F_y = 0 \Rightarrow T - mg = 0 \Rightarrow T = 2 \cdot 9.8 = 19.6\,\text{N}$.

## 2.2 Friction and Springs [Core]

**Concepts and Formulas**

Static friction adjusts to prevent slipping up to its limiting value; kinetic friction opposes sliding. An ideal spring's force points toward equilibrium.

$$\begin{cases} \text{Friction: } f_s \le \mu_s N \text{ (variable up to max)}, \quad f_k = \mu_k N, \\ \text{Spring force: } F = -kx \text{ (Hooke) within elastic limit.} \end{cases}$$

$N$ is the normal force, $\mu_s, \mu_k$ are friction coefficients, $k$ is the spring constant and $x$ is signed displacement from equilibrium.

**Methods and Pitfalls**

- Assume the direction of static friction to be unknown; solve and interpret its sign from the solution.
- Apply $f_s \le \mu_s N$; on the verge of sliding, take $f_{s,\max} = \mu_s N$. For steady sliding, use $f_k = \mu_k N$.
- For springs in series, use $k_{\text{eq}}$: in series $\frac{1}{k_{\text{eq}}} = \sum \frac{1}{k_i}$, in parallel $k_{\text{eq}} = \sum k_i$.

**Example 3.** A 3 kg block on a horizontal surface has $\mu_k = 0.2$. Find the kinetic friction force when the block slides.

**Solution**

$N = mg = 3 \cdot 9.8 = 29.4\,\text{N}$. $f_k = \mu_k N = 0.2 \cdot 29.4 = 5.88\,\text{N}$.

**Example 4.** A spring with constant $k = 200\,\text{N/m}$ is compressed by $x = 0.1\,\text{m}$. Find the restoring force.

**Solution**

The restoring force has magnitude $k|x| = 200 \cdot 0.1 = 20\,\text{N}$ and points opposite to the compression.

## 2.3 Non-uniform Circular Motion Core

**Concepts and Formulas**

For a path of instantaneous radius $r$, choose the radial positive direction toward the center. The tangential component changes speed.

$$\Big\{\text{Radial: } \sum F_r = mv^2/r \;;\; \text{tangential: } \sum F_t = ma_t.$$

**Methods and Pitfalls**

Split radial and tangential components explicitly; centripetal is net radial.

**Example 5.** A car of mass $m$ rounds a curve of radius $r = 50\,\text{m}$ at $v = 20\,\text{m/s}$. Find the required centripetal force.

**Solution**

$$F_c = \frac{mv^2}{r} = m\frac{400}{50} = 8m\,\text{N}.$$

**Example 6.** A ball on a string swings in a vertical circle of radius $r$. At the top, the tension is $T$ and speed is $v$. Write the centripetal force equation.

**Solution**

At the top, $T + mg = \dfrac{mv^2}{r}$ (both point inward).

## 2.4 Impulse and Variable Forces [Extension]

**Concepts and Formulas**

Impulse is the net force integrated over a time interval; it equals the change in momentum.

$$\left\{\text{Impulse: } \vec{J} = \int \vec{F}\, dt = \Delta\vec{p}.\right.$$

**Methods and Pitfalls**

Extrapolate force-time profiles for varying forces; examine external impulses for system imparted momentum changes.

**Example 7.** A constant force $F = 10\,\text{N}$ acts on a mass for $\Delta t = 3\,\text{s}$. Find the impulse delivered.

**Solution**

$J = F\Delta t = 10 \cdot 3 = 30\,\text{N}\cdot\text{s}$.

**Example 8.** A particle's momentum changes from $\vec{p}_i = (5, 0)\,\text{kg}\cdot\text{m/s}$ to $\vec{p}_f = (5, 10)\,\text{kg}\cdot\text{m/s}$ in $2\,\text{s}$. Find the average force.

**Solution**

$\vec{J} = \Delta\vec{p} = (0, 10)\,\text{kg}\cdot\text{m/s}$. $\vec{F}_{\text{avg}} = \dfrac{\vec{J}}{\Delta t} = (0, 5)\,\text{N}$.

**Related problems:** Problem 2 (p. 2), Problem 5 (p. 5), Problem 14 (p. 11), Problem 16 (p. 13), Problem 23 (p. 18), Problem 27 (p. 21), Problem 28 (p. 22), Problem 36 (p. 27)

# 3 Work, Energy, Power

*Coverage:* Work $W = \int \vec{F}\cdot d\vec{r}$; Work–Energy theorem; potential energies (gravity, spring); mechanical energy conservation; power (instantaneous vs average); conservative fields ($F_x = -\,dU/dx$)

## 3.1 Work and the Work–Energy Theorem [Core]

### Concepts and Formulas

Work transfers energy through a force acting over a displacement; the net work equals the change in kinetic energy.

$$\begin{cases} \text{For constant } F \text{ and displacement } d \text{ at angle } \theta: \quad W = Fd\cos\theta, \\ \text{Work–Energy: } \Delta K = W_{\text{net}}, \\ \text{Power: instantaneous } P = \vec{F}\cdot\vec{v}, \quad \text{average over } \Delta t: \ \bar{P} = \dfrac{W}{\Delta t}. \end{cases}$$

### Methods and Pitfalls

- Choose the system first. Account for energy transferred across its boundary and for internal energy changes within it.
- Include nonconservative work explicitly (friction, thrust). For rolling without slipping, static friction often does no work on the rolling body.
- Check for indications by comparing the endpoints: $K_i + U_i$ vs. $K_f + U_f$.

**Example 1.** A 10 kg block is pushed 5 m horizontally by a constant force $F = 30$ N in the direction of motion. Find the work done by that force.

**Solution**

$W = Fd\cos\theta = 30 \cdot 5 \cdot \cos 0 = 150$ J.

**Example 2.** A net force of 50 N accelerates a 5 kg mass from rest over 10 m. Find the final kinetic energy and speed.

**Solution**

$W_{\text{net}} = 50 \cdot 10 = 500$ J. By work-energy theorem, $\Delta K = W_{\text{net}} = 500$ J. Then $\frac{1}{2}mv^2 = 500 \Rightarrow v = \sqrt{200} = 10\sqrt{2}$ m/s.

## 3.2 Potential Energy and Conservation [Core]

### Concepts and Formulas

When nonconservative work is zero, kinetic plus potential energy is conserved.

$$\begin{cases} U_g = mgh \text{ (near-Earth)}, \quad U_s = \frac{1}{2}kx^2, \\ \text{Conservation (when nonconservative work zero): } K_i + U_i = K_f + U_f. \end{cases}$$

### Methods and Pitfalls

Use reference $U = 0$ conveniently; only differences matter.

**Example 3.** A 2 kg block falls $h = 5\,\text{m}$ from rest. Find its speed at the bottom using energy conservation.

### Solution

$K_i + U_i = K_f + U_f \Rightarrow 0 + mgh = \frac{1}{2}mv^2 + 0 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \cdot 5} = \sqrt{98} \approx 9.9\,\text{m/s}$.

**Example 4.** A spring ($k = 100\,\text{N/m}$) is compressed by $x = 0.2\,\text{m}$. Compute the stored elastic potential energy.

### Solution

$U_s = \frac{1}{2}kx^2 = \frac{1}{2} \cdot 100 \cdot 0.04 = 2\,\text{J}$.

## 3.3 Conservative Fields [Extension]

### Concepts and Formulas

In one dimension, a conservative force is the negative spatial derivative of its potential energy:

$$F_x = -\frac{dU}{dx}$$

Path independence in conservative fields; use energy methods when applicable.

### Methods and Pitfalls

Use energy methods if forces are conservative. In a simply connected region, a continuously differentiable force field with zero curl is conservative.

**Example 5.** A mass moves in a potential $U(x) = \frac{1}{2}kx^2$. Find the force acting on it.

**Solution**

$\vec{F} = -\dfrac{dU}{dx}\,\hat{\imath} = -kx\,\hat{\imath}$ (Hooke's law in 1D).

**Example 6.** Show that gravitational potential $U = -\dfrac{GMm}{r}$ gives force $F = -\dfrac{GMm}{r^2}$ (radial).

**Solution**

First compute $\dfrac{dU}{dr} = \dfrac{d}{dr}\Big(-GMm\,r^{-1}\Big) = +\dfrac{GMm}{r^2}$. Therefore the radial force is $\vec{F} = -\dfrac{dU}{dr}\,\hat{\mathbf{r}} = -\dfrac{GMm}{r^2}\,\hat{\mathbf{r}}$ (attractive, inward).

**Related problems:** Problem 2 (p. 2), Problem 14 (p. 11), Problem 17 (p. 13), Problem 20 (p. 16), Problem 28 (p. 22)

# 4 Momentum & Collisions

*Coverage:* Momentum $\vec{p} = m\vec{v}$; impulse; momentum conservation; elastic/inelastic/fully inelastic collisions; center of mass; rocket equation

## 4.1 Momentum and Impulse [Core]

**Concepts and Formulas**

**Concept explanation:** Momentum measures motion; impulse is the total effect of force over time and is equal to the change in momentum.
**Core formulas:**

$$\begin{cases} \vec{p} = m\,\vec{v}, \\ \vec{J} = \displaystyle\int_{t_1}^{t_2} \vec{F}\,dt = \Delta\vec{p}, \\ \text{System:}\quad \vec{P} = \displaystyle\sum_i m_i\vec{v}_i, \quad \Delta\vec{P} = \vec{J}_{\text{ext}}.\ (\text{If } \vec{J}_{\text{ext}} = 0,\ \vec{P} \text{ conserved}) \end{cases}$$

**Variable definitions:** $\vec{p}$ momentum; $\vec{P}$ total momentum; $\vec{J}$ impulse; $\vec{F}$ external force; $m$ mass; $\vec{v}$ velocity.
**Prerequisites & scope:** Valid for Newtonian mechanics; for varying mass systems, take care with momentum flux (see rockets).

**Methods and Pitfalls**

- Identify a closed system (no external impulse) to use momentum conservation immediately.

- For short-duration large forces (collisions), use impulse–momentum rather than force–time details.

- Treating internal forces as external. Fix: define the system to include interacting bodies so internal forces cancel.

**Example 1.** A 4 kg object moves with velocity $\vec{v} = (6, 8)$ m/s. Find its momentum vector and magnitude.

**Solution**

$\vec{p} = m\vec{v} = 4(6, 8) = (24, 32)\,\text{kg} \cdot \text{m/s}$, so $|\vec{p}| = \sqrt{24^2 + 32^2} = 40\,\text{kg} \cdot \text{m/s}$ and the direction matches $\vec{v}$.

**Example 2.** A force $F(t) = 10t$ N acts on a particle from $t = 0$ to $t = 2$ s. Find the impulse delivered.

**Solution**

$J = \int_0^2 10t\,dt = [5t^2]_0^2 = 20\,\text{N} \cdot \text{s}$.

## 4.2 Elastic and Inelastic Collisions [Core]

**Concepts and Formulas**

**Concept explanation:** When the net external impulse is negligible, collisions conserve total momentum; elastic ones also conserve kinetic energy. The center-of-mass (COM) frame makes algebra simpler.
**Core formulas (1D):**

$$\begin{cases} \text{Momentum: } m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2, \\ \text{Elastic energy: } \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2, \\ \text{Result (elastic): } v_1 = \dfrac{(m_1 - m_2)u_1 + 2m_2u_2}{m_1 + m_2}, \quad v_2 = \dfrac{2m_1u_1 + (m_2 - m_1)u_2}{m_1 + m_2}, \\ \text{Completely inelastic: stick } (v_1 = v_2 = v) = \dfrac{m_1u_1 + m_2u_2}{m_1 + m_2}. \end{cases}$$

**Variable definitions:** $u_i$ initial, $v_i$ final velocities; $m_i$ masses; all along one line (1D).
**Prerequisites & scope:** For oblique/2D, conserve vector momentum and use geometry; kinetic energy changes via deformation/heat in inelastic cases.

**Methods and Pitfalls**

- Switch to the COM frame where total momentum is zero; velocities reverse in elastic 1D collisions.

- For 2D glancing collisions, conserve components along orthogonal axes; use

restitution or geometry for angles.

- Enforcing kinetic energy conservation in inelastic impact. Fix: only momentum is guaranteed; account for energy loss.

**Example 3.** Two carts collide elastically in 1D. Mass $m_1 = 2\,\text{kg}$ at $u_1 = 5\,\text{m/s}$ hits mass $m_2 = 3\,\text{kg}$ at rest. Find $v_1$ and $v_2$.

**Solution**

$v_1 = \dfrac{(2-3)\cdot 5 + 2\cdot 3\cdot 0}{5} = \dfrac{-5}{5} = -1\,\text{m/s}.$ $v_2 = \dfrac{2\cdot 2\cdot 5 + (3-2)\cdot 0}{5} = \dfrac{20}{5} = 4\,\text{m/s}.$

**Example 4.** In a perfectly inelastic collision, $m_1 = 4\,\text{kg}$ at $u_1 = 6\,\text{m/s}$ collides with $m_2 = 2\,\text{kg}$ at rest. Find the final velocity.

**Solution**

$$v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2} = \frac{4\cdot 6 + 0}{6} = 4\,\text{m/s}.$$

## 4.3 Center of Mass and System Dynamics [Core]

**Concepts and Formulas**

**Conceptual explanation:** The center of mass (COM) is an average of the mass distribution of a system; external forces accelerate the COM as if all the mass were centered at the COM.
**Core formulas:**

$$\begin{cases} \vec{R} = \dfrac{\sum m_i \vec{r}_i}{\sum m_i}, \quad \vec{V} = \dot{\vec{R}} = \dfrac{\sum m_i \vec{v}_i}{M}, \\ M\,\ddot{\vec{R}} = \sum \vec{F}_{\text{ext}}. \text{ (Internal forces cancel in pairs)} \end{cases}$$

**Variable definitions:** $\vec{r}_i, \vec{v}_i$ positions/velocities; $M = \sum m_i$ total mass; $\vec{R}$ COM position.
**Prerequisites & scope:** Requires Newton's third law in internal pairs; for variable mass, include momentum flux.

**Methods and Pitfalls**

- Compute COM motion to track overall translation, then analyze internal relative motion separately.
- In explosions/fragmentation, the COM continues with pre-event velocity if external forces are negligible.

- Treating internal impulses as changing COM momentum. Fix: only external impulse changes total momentum.

**Example 5.** Two masses $m_1 = 3\,\text{kg}$ at $x_1 = 0$ and $m_2 = 2\,\text{kg}$ at $x_2 = 5\,\text{m}$ lie on a line. Find the COM position.

**Solution**

$$X = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} = \frac{0 + 10}{5} = 2\,\text{m}.$$

**Example 6.** A system has two equal masses $m$ moving at velocities $\vec{v}_1 = (2, 0)$ and $\vec{v}_2 = (-1, 3)$ m/s. Find the velocity of the COM.

**Solution**

$$\vec{V} = \frac{m\vec{v}_1 + m\vec{v}_2}{2m} = \frac{(1, 3)}{2} = (0.5, 1.5)\,\text{m/s}.$$

## 4.4 Variable Mass and Rockets **Extension**

**Concepts and Formulas**

**Concept explanation:** For mass-exchange systems (rockets), momentum conservation for rocket+exhaust yields logarithmic change in velocity.
**Core formulas:**

$$\begin{cases} \text{Continuous: } m\,d\vec{v} = \vec{v}_e\,dm, \\ \text{where } dm < 0 \text{ and } \vec{v}_e \text{ is the backward exhaust velocity relative to the rocket,} \\ \text{Ideal Tsiolkovsky (1D): } \Delta v = v_e \ln \dfrac{m_i}{m_f}. \end{cases}$$

**Variable definitions:** $\vec{v}_e$ is the backward exhaust velocity relative to the rocket; $v_e = |\vec{v}_e|$ is its positive magnitude. $m_i, m_f$ are initial/final mass and $m$ is instantaneous mass.
**Prerequisites & scope:** Neglect external forces (or include gravity drag separately); $v_e$ constant; exhaust ejected at steady relative speed.

**Methods and Pitfalls**

- Include gravity losses by subtracting $g\,\Delta t$ from $\Delta v$ when appropriate (vertical ascent approximation).
- Use staging by summing $v_e \ln(m_i/m_f)$ per stage.
- Using exhaust speed relative to Earth instead of rocket. Fix: $v_e$ is defined relative to the rocket.

**Example 7.** A rocket has initial mass $m_i = 1000\,\text{kg}$, final mass $m_f = 400\,\text{kg}$, and exhaust speed $v_e = 2000\,\text{m/s}$. Find $\Delta v$ in space (ignoring gravity).

**Solution**

$$\Delta v = v_e \ln \frac{m_i}{m_f} = 2000 \ln \frac{1000}{400} = 2000 \ln 2.5 \approx 1833\,\text{m/s}.$$

**Example 8.** A rocket ejects mass at rate $\dot{m} = 10\,\text{kg/s}$ with effective exhaust speed $v_e = 1500\,\text{m/s}$. Find the instantaneous thrust force.

**Solution**

$$\text{Thrust} = \dot{m}\, v_e = 10 \cdot 1500 = 15000\,\text{N}.$$

**Related problems:** Problem 1 (p. 1), Problem 29 (p. 23), Problem 33 (p. 25)

# 5 Rotational Motion

*Coverage:* Angular kinematics; torque ($|\boldsymbol{\tau}| = rF\sin\theta$); moment of inertia; $\tau_{\text{net}} = I\alpha$; rotational kinetic energy; angular momentum and conservation; rolling without slipping; inertia integrals

## 5.1 Angular Kinematics and Dynamics Core

**Concepts and Formulas**

**Concept description:** Rotational motion is analog to linear motion: torque is the rotational analog to force, and moment of inertia of mass.
**Core formulas:**

$$\begin{cases} \omega = \dfrac{d\theta}{dt}, \quad \alpha = \dfrac{d\omega}{dt}, \text{ with constant-}\alpha\text{ kinematics.} \\ \boldsymbol{\tau} = \vec{r} \times \vec{F}, \quad |\boldsymbol{\tau}| = r_\perp F = rF\sin\theta, \quad \sum \tau = I\alpha, \\ K_r = \frac{1}{2} I\omega^2, \quad P = \tau\,\omega. \end{cases}$$

**Variable definitions:** $\theta, \omega, \alpha$ angular position/velocity/acceleration; $\tau$ torque; $I$ moment of inertia.
**Prerequisites & scope:** Rigid body about a fixed axis; $I$ constant in time.

**Methods and Pitfalls**

- Use the perpendicular lever arm $r_\perp$ for torques; sum about convenient pivots to kill unknown forces.
- Prefer energy when forces are complicated but conservative; otherwise use $\sum \tau = I\alpha$ about the COM or a fixed axis.
- Mixing signs of torques from different reference senses. Fix: choose a positive rotation sense and stick with it.

**Example 1.** A disk of radius $r = 0.5\,\mathrm{m}$ and moment of inertia $I = 2\,\mathrm{kg}\cdot\mathrm{m}^2$ experiences a net torque $\tau = 10\,\mathrm{N}\cdot\mathrm{m}$. Find its angular acceleration.

**Solution**

$\alpha = \dfrac{\tau}{I} = \dfrac{10}{2} = 5\,\mathrm{rad/s}^2$.

**Example 2.** A flywheel rotating at $\omega = 20\,\mathrm{rad/s}$ has $I = 5\,\mathrm{kg}\cdot\mathrm{m}^2$. Find its rotational kinetic energy.

**Solution**

$K_r = \frac{1}{2}I\omega^2 = \frac{1}{2}\cdot 5\cdot 400 = 1000\,\mathrm{J}$.

## 5.2 Angular Momentum and Conservation [Core]

**Concepts and Formulas**

**Explanation of concept:** Angular momentum is conserved when no external torque is applied.
**Core formulas:**

$$\begin{cases} L_{\text{axis}} = I\omega \text{ (about a fixed axis)}, & \sum \vec{\tau}_{\text{ext}} = \dfrac{d\vec{L}}{dt}. \\ \text{If } \sum \tau_{\text{ext}} = 0,\ \vec{L} \text{ conserved.} \end{cases}$$

**Variable definitions:** $\vec{L}$ angular momentum; $I$ moment of inertia.
**Prerequisites & scope:** Axis and point of reference must be specified; rolling applies at instantaneous point of contact.

**Methods and Pitfalls**

- For isolated systems with negligible external torques, apply $\vec{L}$ conservation about a fixed axis.
- Choose the reference point wisely to eliminate unknown torques.
- Forgetting that static friction can act either way in rolling. Fix: determine its direction from torque/acceleration requirements.

**Example 3.** A disk of radius $R$ rolls without slipping at $v_{cm} = 5\,\text{m/s}$. Find $\omega$.

**Solution**

$$v_{cm} = \omega R \Rightarrow \omega = \frac{v_{cm}}{R} = \frac{5}{R}\,\text{rad/s}.$$

**Example 4.** A solid sphere ($I = \frac{2}{5}mR^2$) rolls without slipping at $\omega = 10\,\text{rad/s}$ and $v_{cm} = 2\,\text{m/s}$. Find its total kinetic energy if $m = 3\,\text{kg}$.

**Solution**

Using no-slip $v_{cm} = \omega R$, we have $I\omega^2 = \frac{2}{5}mR^2\omega^2 = \frac{2}{5}mv_{cm}^2$. Thus $K = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I\omega^2 = \left(\frac{1}{2} + \frac{1}{5}\right)mv_{cm}^2 = \frac{7}{10}\,mv_{cm}^2 = \frac{7}{10} \cdot 3 \cdot 4 = 8.4\,\text{J}$.

## 5.3 Moments of Inertia **Extension**

### Concepts and Formulas

**Concept description:** Moment of inertia is a measure of resistance to angular acceleration; composition rules and standard shapes allow for rapid calculation.
**Core formulas:**

$$\begin{cases} I = \int r^2\,dm \text{ (axis distance } r), \\ \text{Parallel axis: } I = I_{\text{cm}} + Md^2, \\ \text{Perpendicular axis (planar lamina): } I_z = I_x + I_y. \end{cases}$$

**Variable definitions:** $I_{\text{cm}}$ about COM axis; $d$ offset; $M$ total mass.
**Prerequisites & scope:** Perpendicular-axis requires lamina in the plane; parallel-axis requires fixed, parallel axes.

### Methods and Pitfalls

- Decompose into standard shapes and sum moments about the same axis.
- Use symmetry to eliminate products of inertia; choose axes through COM when possible.
- Applying perpendicular-axis to 3D bodies. Fix: valid only for planar laminae.

**Example 5.** A thin rod of mass $m$ and length $L$ rotates about one end. Find its moment of inertia.

**Solution**

$I = \int_0^L x^2\,dm = \int_0^L x^2 \frac{m}{L}\,dx = \frac{m}{L}\frac{L^3}{3} = \frac{mL^2}{3}$.

**Example 6.** A disk of mass $M$ and radius $R$ has $I_{cm} = \frac{1}{2}MR^2$. Using the parallel-axis theorem, find $I$ about a point on its rim.

**Solution**

$I = I_{cm} + MR^2 = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$.

## 5.4 Rolling Without Slipping Core

**Concepts and Formulas**

**Concept description:** Rolling is a mixture of translation of the center of mass and rotation; the no-slip condition ties them together.
**Core formulas:**

$$\begin{cases} v_{\text{cm}} = \omega R, \\ K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I\omega^2. \end{cases}$$

**Variable definitions:** $R$ radius; $M$ mass; $I$ moment of inertia; $v_{\text{cm}}$ center-of-mass speed.

**Methods and Pitfalls**

- Static friction can accelerate or decelerate a rolling body. For a rigid body rolling without slipping on a stationary surface, the contact point is instantaneously at rest and static friction does no work on the body.
- Use energy for ramp problems; use $\sum \tau = I\alpha$ when forces/accelerations are requested.

**Example 7.** A solid cylinder of mass $m$ and radius $R$ is released from rest to roll without slipping down an incline of height $h$. Find its speed at the bottom.

**Solution**

Energy: $mgh = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}(\frac{1}{2}mR^2)(v_{cm}^2/R^2) = \frac{1}{2}mv_{cm}^2 + \frac{1}{4}mv_{cm}^2 = \frac{3}{4}mv_{cm}^2$, so $v_{cm} = \sqrt{\frac{4}{3}gh}$.

**Example 8.** A solid sphere rolls without slipping down a rough incline under gravity. Determine the direction of static friction on the sphere.

**Solution**

Gravity acts through the center, so static friction supplies the torque needed for rolling. It points up the incline, producing rotation in the direction of motion. In a different driven-rolling situation the friction direction can differ; velocity alone does not determine it.

**Related problems:** Problem 15 (p. 12), Problem 16 (p. 13), Problem 20 (p. 16), Problem 38 (p. 29)

# Part II: Oscillations, Waves, Thermodynamics & Fluids

# 6 Oscillations & Waves

*Coverage:* SHM; pendulum (small-angle); wave speed $v = f\lambda$; superposition/standing waves; sound and Doppler; SHM ODE

## 6.1 Simple Harmonic Motion [Core]

**Concepts and Formulas**

**Concept explanation:** SHM occurs when acceleration is proportional to and opposite to displacement; solutions are sinusoidal with constant amplitude (no damping).
**Core formulas:**

$$\begin{cases} x = A\cos(\omega t + \phi), \quad v = -A\omega\sin(\omega t + \phi), \quad a = -\omega^2 x, \\ T = \dfrac{2\pi}{\omega}, \quad \omega = \sqrt{\dfrac{k}{m}} \text{ (mass–spring)}, \quad T_{\text{pend}} \approx 2\pi\sqrt{\dfrac{\ell}{g}} \text{ (small angle)}. \end{cases}$$

**Variable definitions:** $A$ amplitude; $\omega$ angular frequency; $\phi$ phase; $k$ spring constant; $\ell$ pendulum length.
**Prerequisites & scope:** No damping/driving; small-angle approximation for pendulum.

**Methods and Pitfalls**

- Use energy partition $K + U = \frac{1}{2}kA^2$ to find speeds at positions; use phase to compute time fractions.
- For compound oscillators, reduce to effective $k_{\text{eff}}$ or $\ell_{\text{eff}}$ before applying SHM formulas.
- Small-angle pendulum: check $\theta_{\max} \lesssim 10^\circ$ for $T \approx 2\pi\sqrt{\ell/g}$ to be within a percent; otherwise expect longer $T$. First-order correction (radians; [Extension] result,

memorize only): $T \approx 2\pi\sqrt{\ell/g}\left(1+\frac{\theta_0^2}{16}\right)$.

- Using pendulum period formula at large angles. Fix: restrict to small angles or use elliptic corrections.

**Example 1.** A mass on a spring follows SHM with amplitude $A$ and period $T$. What fraction of the period is spent with $|x| > \frac{A}{2}$?

**Solution**

Let $x(t) = A\cos(\omega t)$. The condition $|x(t)| > A/2$ is equivalent to $|\cos(\omega t)| > 1/2$. It is simpler to calculate the fraction of time for the complementary condition, $|x(t)| \le A/2$, which corresponds to $|\cos(\omega t)| \le 1/2$. In one full cycle $\theta = \omega t \in [0, 2\pi)$, this holds for $\theta \in [\pi/3, 2\pi/3]$ and $\theta \in [4\pi/3, 5\pi/3]$. The total angular duration is $(\frac{2\pi}{3} - \frac{\pi}{3}) + (\frac{5\pi}{3} - \frac{4\pi}{3}) = \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$. The fraction of the period for this condition is $\frac{2\pi/3}{2\pi} = \frac{1}{3}$. Therefore, the fraction of the period spent with $|x| > A/2$ is $1 - 1/3 = 2/3$.

## 6.2 Waves (Traveling) Core

**Concepts and Formulas**

**Concept explanation:** Traveling waves follow $v = f\lambda$; boundary conditions set standing-wave modes; source/observer motion shifts frequency (Doppler).
**Core formulas:**

$$\Big\{v = f\lambda, \quad y(x,t) = A\cos(kx - \omega t + \phi), \ k = \tfrac{2\pi}{\lambda}, \ \omega = 2\pi f.$$

**Variable definitions:** $v$ wave speed in medium; $L$ length; $v_o$ observer speed; $v_s$ source speed.
**Prerequisites & scope:** Linear superposition; small amplitudes; Doppler formula assumes $v_o, v_s \ll v$ (nonrelativistic).

**Methods and Pitfalls**

- Draw mode shapes to match node/antinode boundary conditions before writing $f_n$.
- Use a sign diagram for Doppler to avoid sign errors; approaching increases frequency. For the formula given, this means observer towards source ($+v_o$, numerator) and source towards observer ($-v_s$, denominator). For receding, reverse these signs.
- Using $v$ of sound/light incorrectly across media. Fix: use the correct medium speed for $v = f\lambda$.

**Example 2.** A wave has wavelength $\lambda = 2\,\text{m}$ and frequency $f = 50\,\text{Hz}$. Find its speed.

**Solution**

$v = f\lambda = 50 \cdot 2 = 100\,\text{m/s}$.

## 6.3 Standing Waves (Strings/Pipes) Core

**Concepts and Formulas**

**Concept explanation:** Boundaries reflect waves; interference of forward and backward waves creates nodes and antinodes with discrete mode frequencies set by geometry and boundary conditions. **Core formulas:**

$$\begin{cases} \text{String fixed ends: } f_n = \dfrac{nv}{2L},\ n = 1, 2, \ldots \\ \text{Open/closed pipe: } \begin{cases} f_n = \dfrac{nv}{2L}, & n = 1, 2, 3, \ldots \text{ (both ends open)}, \\ f_n = \dfrac{(2n-1)v}{4L}, & n = 1, 2, 3, \ldots \text{ (one end closed)}. \end{cases} \end{cases}$$

**Example 3.** A string of length $L = 1.2\,\text{m}$ fixed at both ends has fundamental frequency $f_1 = 200\,\text{Hz}$. Find the wave speed.

**Solution**

$f_1 = \dfrac{v}{2L} \Rightarrow v = 2Lf_1 = 2 \cdot 1.2 \cdot 200 = 480\,\text{m/s}$.

## 6.4 Doppler Effect (Fixed Medium) Core

**Concepts and Formulas**

**Concept explanation:** Relative motion between source and observer shifts the detected frequency: approaching raises $f'$ and receding lowers it; the medium is stationary. **Core formula:**

$$f' = \frac{v \pm v_o}{v \mp v_s} f \text{ (approach uses top signs).}$$

**Variable definitions:** $v$ wave speed; $v_o$ observer speed; $v_s$ source speed.

**Example 4.** An acoustic source emits $f = 440\,\text{Hz}$ in air with $v = 340\,\text{m/s}$. The observer moves toward the source at $v_o = 10\,\text{m/s}$ while the source moves toward the observer at $v_s = 20\,\text{m/s}$. Find the observed frequency.

**Solution**

$$f' = \frac{v + v_o}{v - v_s} f = \frac{350}{320} \cdot 440 \approx 481\,\text{Hz}.$$

## 6.5 SHM ODE and Driven Forms [Extension]

**Concepts and Formulas**

**Concept explanation:** The homogeneous SHM ODE has sinusoidal solutions; with driving and damping, the steady-state response depends on drive frequency and damping ratio.
**Core formulas:**

$$\begin{cases} x'' + \omega_0^2 x = 0 \Rightarrow x(t) = C\cos\omega_0 t + D\sin\omega_0 t, \\ \text{Damped: } x'' + 2\zeta\omega_0 x' + \omega_0^2 x = 0, \text{ under/critical/over-damped by } \zeta, \\ \text{Driven: } x'' + 2\zeta\omega_0 x' + \omega_0^2 x = \dfrac{F_0}{m}\cos\omega t, \\ \Rightarrow \text{ amplitude peaks near } \omega \approx \omega_0, \\ \text{for small damping } \omega_{\text{peak}} \approx \omega_0\sqrt{1-2\zeta^2}\ (\zeta \ll 1). \end{cases}$$

**Variable definitions:** $\omega_0$ natural frequency; $\zeta$ damping ratio; $F_0$ drive amplitude.
**Prerequisites & scope:** Linear oscillator model; small oscillations; steady-state assumes transients have decayed.

**Methods and Pitfalls**

- Identify regime via $\zeta$; near resonance, estimate amplification and phase shift.
- Confusing natural and driving frequencies. Fix: keep $\omega_0$ (system) distinct from $\omega$ (drive).

**Example 5.** For $x'' + \omega^2 x = 0$ with $x(0) = 0$ and $\dot{x}(0) = v_0$, find $x(t)$ and the maximum speed.

**Solution**

The general solution is $x(t) = C\cos(\omega t) + D\sin(\omega t)$. $x(0) = 0 \Rightarrow C = 0$. $\dot{x}(t) = D\omega\cos(\omega t)$, so $\dot{x}(0) = v_0 \Rightarrow D\omega = v_0 \Rightarrow D = v_0/\omega$. Thus, $x(t) = \dfrac{v_0}{\omega}\sin(\omega t)$. The velocity is $v(t) = \dot{x}(t) = v_0\cos(\omega t)$. The maximum speed is the amplitude of $v(t)$, which is $|v_0|$.

**Related problems:** Problem 3 (p. 3), Problem 24 (p. 19), Problem 31 (p. 24)

# 7 Fluids & Thermodynamics

*Coverage:* Hydrostatics (pressure/buoyancy); continuity; Bernoulli; ideal gas; First Law and engines; entropy

## 7.1 Hydrostatics and Buoyancy [Core]

**Concepts and Formulas**

**Concept explanation:** Static fluids exert pressure that increases with depth; the buoyant force equals the weight of displaced fluid (Archimedes).
**Core formulas:**

$$\begin{cases} P = P_0 + \rho g h \text{ (hydrostatic pressure)}, \\ F_b = \rho g V_{\text{disp}} \text{ (buoyancy)}. \end{cases}$$

**Variable definitions:** $P_0$ reference pressure (often atmospheric at $h = 0$); $\rho$ fluid density; $h$ depth; $V_{\text{disp}}$ displaced volume.
**Prerequisites & scope:** Fluid at rest (no flow), constant $\rho$ with depth (or integrate if varying); neglect surface tension unless specified.

**Methods and Pitfalls**

- Draw free-body diagrams (FBD) of floating/sinking bodies: set $F_b$ vs weight vs any tension to solve equilibrium.
- Choose a definite reference level for $h$ and keep $P_0$ the same when comparing points.
- Submerged fraction for floating: $\dfrac{V_{sub}}{V} = \dfrac{\rho_b}{\rho}$ (with $0 < \rho_b \leq \rho$).
- If $\rho_b > \rho$ and the object is released freely, the initial net force is downward: $mg - F_b > 0$.
- If later supported from below or by upward tension, static equilibrium requires $N + T + F_b = mg$, omitting forces that are absent.
- Gauge vs absolute pressure: $\Delta P = \rho g h$ is a *gauge* difference; absolute pressure is $P = P_{\text{atm}} + \rho g h$ when the surface is open to atmosphere.
- Using object's volume instead of displaced volume for $F_b$. Fix: use actual displaced fluid volume (submerged part only).

**Example 1.** A block of volume $V$ and density $\rho_b$ floats in a liquid of density $\rho$. What fraction of its volume is submerged?

**Solution**

At equilibrium $\rho g V_{sub} = \rho_b g V$, so $V_{sub}/V = \rho_b/\rho$.

## 7.2 Continuity and Bernoulli [Core]

**Concepts and Formulas**

**Concept explanation:** In steady incompressible flow, mass conservation gives $Av = \text{const}$; along a streamline with negligible viscosity, mechanical energy per volume is constant (Bernoulli).
**Core formulas:**

$$\begin{cases} \text{Continuity: } A_1 v_1 = A_2 v_2 \ (\rho \text{ constant}). \\ \text{Bernoulli: } P + \frac{1}{2}\rho v^2 + \rho g y = \text{const (along a streamline)}. \end{cases}$$

**Variable definitions:** $A$ cross-sectional area; $v$ speed; $P$ pressure; $y$ elevation; $\rho$ density.
**Prerequisites & scope:** Steady, incompressible, non-viscous flow; apply Bernoulli along a streamline, not across shocks or with pumps/turbines unaccounted.

**Methods and Pitfalls**

- Along a streamline, check steady, incompressible, inviscid flow before using Bernoulli. An irrotational flow also permits the same constant across streamlines.
- Combine continuity with Bernoulli to eliminate speeds or pressures efficiently.
- Use stagnation points: where $v = 0$, the total (stagnation) pressure is $P_0 = P + \frac{1}{2}\rho v^2$ upstream along a streamline.
- Using Bernoulli across different streamlines where viscous losses or pumps exist. Fix: apply along a single streamline and include head gains/losses when needed.

**Example 2.** Water flows through a horizontal pipe from diameter $D_1 = 0.1\,\text{m}$ to $D_2 = 0.05\,\text{m}$. If $v_1 = 2\,\text{m/s}$, find $v_2$ using continuity.

**Solution**

$$A_1 v_1 = A_2 v_2 \Rightarrow \pi(D_1/2)^2 v_1 = \pi(D_2/2)^2 v_2 \Rightarrow v_2 = v_1 \frac{D_1^2}{D_2^2} = 2\frac{0.01}{0.0025} = 8\,\text{m/s}.$$

**Example 3.** At point 1 in a pipe, $P_1 = 1.0 \times 10^5\,\text{Pa}$, $v_1 = 2\,\text{m/s}$, $y_1 = 0$. At point 2, $v_2 = 5\,\text{m/s}$, $y_2 = 3\,\text{m}$. Find $P_2$ for water ($\rho = 1000\,\text{kg/m}^3$).

**Solution**

Bernoulli: $P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$. Compute in SI with scientific notation: $\frac{1}{2}\rho(v_1^2 - v_2^2) = 0.5 \times 10^3\,(4 - 25) = -1.05 \times 10^4\,\text{Pa}$ and $\rho g(y_2 - y_1) = 10^3 \times 9.8 \times 3 = 2.94 \times 10^4\,\text{Pa}$. Hence $P_2 = 1.00 \times 10^5 - 1.05 \times 10^4 - 2.94 \times 10^4 = 6.01 \times 10^4\,\text{Pa}$.

## 7.3 Ideal Gas and First Law Core

**Concepts and Formulas**

**Concept explanation:** $PV = nRT$ holds for ideal gases; the First Law links changes in internal energy to heat and work with clear sign conventions.
**Core formulas:**

$$\begin{cases} PV = nRT, \quad U = \frac{f}{2}nRT, \\ \text{where } f \text{ is dof (e.g., } f = 3 \text{ monatomic, } 5 \text{ diatomic at room T)}, \\ \Delta U = Q + W_{\text{on}} \quad (W_{\text{on}} = \text{work done on the system}), \\ W_{\text{on}} = -\int P_{\text{ext}}\,dV, \quad W_{\text{by}} = -W_{\text{on}}. \end{cases}$$

**Variable definitions:** $P, V, T$ pressure/volume/temperature; $n$ moles; $R$ gas constant; $Q$ heat into system; $W_{\text{on}}$ work on system; $W_{\text{by}}$ work done by the gas; $U$ internal energy.
**Prerequisites & scope:** Ideal gas approximation; $U$ depends only on $T$ for ideal gases. The stated $f$ values use classical equipartition. For quasistatic expansion $P_{\text{ext}} = P$; in general use the external pressure for boundary work.

**Methods and Pitfalls**

- Identify process (isochoric/isobaric/isothermal/adiabatic) to pick $W$, $Q$, $\Delta U$ quickly.
- Draw $P$–$V$ diagrams: areas give work; direction indicates sign.
- Mixing sign conventions for work. Fix: adopt $\Delta U = Q + W_{\text{on}}$ (work on system positive) consistently; then $W_{\text{by}} = -W_{\text{on}}$.

**Example 4.** An ideal gas undergoes an isothermal expansion from volume $V_1$ to $V_2$ at temperature $T$. Find the work done by the gas.

**Solution**

$W = \int_{V_1}^{V_2} \frac{nRT}{V}\, dV = nRT \ln \frac{V_2}{V_1}$.

**Example 5.** An ideal gas at $P = 2 \times 10^5\,\text{Pa}$ and $V = 0.01\,\text{m}^3$ has $n = 1\,\text{mol}$. Find the temperature $T$ using $PV = nRT$ with $R = 8.314\,\text{J/(mol}\cdot\text{K)}$.

**Solution**

$$T = \frac{PV}{nR} = \frac{2 \times 10^5 \cdot 0.01}{1 \cdot 8.314} \approx 240\,\text{K}.$$

## 7.4 Entropy and Carnot [Core]

**Concepts and Formulas**

**Concept explanation:** The total entropy of an isolated system increases in an irreversible process; the entropy of a subsystem can decrease. Carnot gives the maximum efficiency of engines operating between two reservoirs.

**Core formulas:**

$$\begin{cases} \Delta S = \displaystyle\int_{\text{rev}} \frac{\delta Q}{T}, \\ \text{Carnot bound: } \eta_{\max} = 1 - \dfrac{T_c}{T_h}. \end{cases}$$

**Variable definitions:** $S$ entropy; $T_h, T_c$ hot/cold absolute temperatures; $\delta Q$ infinitesimal heat (reversible path).

**Prerequisites & scope:** Absolute temperatures (Kelvin); reversible paths for definition; real engines achieve less than Carnot due to irreversibilities.

**Methods and Pitfalls**

- Compute $\Delta S$ along a convenient reversible path (e.g., isothermal + isochoric steps).
- For engine limits, compare cycle temperatures to $T_h, T_c$ to bound $\eta$ quickly.
- Using Celsius in $\eta = 1 - T_c/T_h$. Fix: convert to Kelvin.

**Example 6.** A Carnot engine operates between $T_h = 500\,\mathrm{K}$ and $T_c = 300\,\mathrm{K}$. Find its maximum efficiency.

**Solution**

$$\eta_{\max} = 1 - \frac{T_c}{T_h} = 1 - \frac{300}{500} = 0.4 = 40\%.$$

**Example 7.** An ideal gas expands reversibly at constant $T = 400\,\mathrm{K}$ from $V_1 = 1\,\mathrm{m}^3$ to $V_2 = 2\,\mathrm{m}^3$. Find the entropy change if $n = 1\,\mathrm{mol}$.

**Solution**

$$\Delta S = \frac{Q_{rev}}{T} = \frac{nRT\ln(V_2/V_1)}{T} = nR\ln 2 = 8.314\ln 2 \approx 5.76\,\mathrm{J/K}.$$

## 7.5 Heat Engines and Efficiency [Core]

**Concepts and Formulas**

**Core formulas:**

$$\begin{cases} W_{\mathrm{by}} = \displaystyle\int P_{\mathrm{ext}}\,dV, \quad W_{\mathrm{on}} = -W_{\mathrm{by}}, \quad \eta = \frac{W_{\mathrm{net,by}}}{Q_h}. \end{cases}$$

**Variable definitions:** $Q_h$ is heat absorbed from the hot reservoir; $W_{\mathrm{net,by}}$ is net work per cycle. The Carnot bound is given in the preceding section. For quasistatic steps, $P_{\mathrm{ext}} = P$.

**Related problems:** Problem 4 (p. 4), Problem 6 (p. 5), Problem 11 (p. 9), Problem 18 (p. 14), Problem 19 (p. 15), Problem 37 (p. 28)

# Part III: Electricity & Magnetism

# 8 Electrostatics

*Coverage:* Coulomb force; electric field/potential; capacitors and energy; Gauss's law

## 8.1 Coulomb, Field and Potential [Core]

**Concepts and Formulas**

**Concept explanation:** Point charges interact by an inverse-square law; electric field and potential describe force per unit charge and energy per unit charge.
**Core formulas:**

$$\begin{cases} |\vec{F}| = k\,\dfrac{|q_1 q_2|}{r^2} \text{ (along the line of centers)}, \\ \vec{E} = \dfrac{\vec{F}}{q}, \quad V = \dfrac{U}{q}, \quad \Delta U = -q\displaystyle\int \vec{E}\cdot d\vec{r}, \\ \text{Point charge: } \vec{E} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{r^2}\hat{\mathbf{r}}, \quad V = \dfrac{1}{4\pi\varepsilon_0}\dfrac{Q}{r}. \end{cases}$$

**Variable definitions:** $q, Q$ charges; $r$ separation; $k = 1/(4\pi\varepsilon_0)$; $U$ potential energy.
**Prerequisites & scope:** Electrostatics (charges at rest); superposition holds; signs determine directions.

**Methods and Pitfalls**

- Apply symmetry (dipoles, rings, infinite sheets) to cancel parts before integrating.
- Apply potential for conservative additions first, then differentiate to get fields.
- Not remembering vector directions for $\vec{E}$ and $\vec{F}$. Fix: graph direction first, calculate magnitude second.

**Example 1.** Two point charges $+Q$ are at $(\pm a, 0)$. Find the electric field on the $y$-axis at $(0, y)$.

**Solution**

Horizontal components cancel; vertical add: $E_y = 2\,kQ\,\dfrac{y}{(a^2+y^2)^{3/2}}$.

**Key Insight**

**Constants/units:** $\varepsilon_0 = 8.85\times10^{-12}\,\mathrm{F/m}$, $\mu_0 = 4\pi\times10^{-7}\,\mathrm{H/m}$, $c = 3.00\times10^{8}\,\mathrm{m/s}$, $e = 1.60\times10^{-19}\,\mathrm{C}$. Use SI unless specified.

## 8.2 Capacitors and Energy [Core]

### Concepts and Formulas

**Concept explanation:** A capacitor is a device that holds equal and opposite charge on two conductors with a gap or dielectric between them. The stored charge at a given potential difference will be proportional to the geometry and material. Networks simplify by simple series/parallel formulas and energy can be traced by $U = \frac{1}{2}CV^2$.
**Core formulas:**

$$\begin{cases} C = \varepsilon_0 \dfrac{A}{d} \text{ (parallel plates in vacuum)}, \quad C = \varepsilon_r \varepsilon_0 \dfrac{A}{d} \text{ (uniform dielectric)}, \\ \text{Series: } \dfrac{1}{C_s} = \sum \dfrac{1}{C_i}, \quad \text{Parallel: } C_p = \sum C_i, \\ U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \dfrac{Q^2}{2C}. \end{cases}$$

**Variable definitions:** $A$ plate area; $d$ separation; $\varepsilon_0$ vacuum permittivity; $\varepsilon_r$ relative permittivity; $Q$ charge; $V$ voltage.
**Prerequisites & scope:** Edge effects neglected; linear dielectrics; use equivalent capacitance to reduce networks.

### Methods and Pitfalls

- Use series and parallel combinations or symmetry to simplify a network before writing node or loop equations.
- Use $U = \frac{1}{2}CV^2$ to compare energy storage or redistribution after reconfiguration.
- Assuming charge conservation on each plate when switches change connectivity. Fix: conserve charge on isolated conductors only.

**Example 2.** Two capacitors $C_1 = 2\,\mu\text{F}$ and $C_2 = 3\,\mu\text{F}$ are in series. Find the equivalent capacitance.

**Solution**

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6} \Rightarrow C_s = \frac{6}{5} = 1.2\,\mu\text{F}.$$

**Example 3.** A capacitor with $C = 10\,\mu\text{F}$ is charged to $V = 50\,\text{V}$. Find the energy stored.

**Solution**

$U = \frac{1}{2}CV^2 = \frac{1}{2} \cdot 10 \times 10^{-6} \cdot 2500 = 12.5 \times 10^{-3} = 12.5\,\text{mJ}$.

## 8.3 Gauss's Law [Core]

**Concepts and Formulas**

**Concept explanation:** The flux of $\vec{E}$ through a closed surface equals enclosed charge over $\varepsilon_0$; symmetry lets you get fields without integration.
**Core formula:**

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\varepsilon_0}.$$

**Variable definitions:** $Q_{enc}$ charge enclosed; $d\vec{A}$ outward area element; choose Gaussian surface aligned to symmetry.
**Prerequisites & scope:** Use for infinite planes/cylinders/spheres; for conductors, $E = 0$ inside and charges reside on surfaces. Within uniform dielectrics/insulators with embedded charge, fields may exist inside the material (i.e., $E$ need not vanish).
*Under [Core]: memorize the integral statement; derivations are not required.*

**Methods and Pitfalls**

- Pick surfaces where $E$ is constant and parallel to $d\vec{A}$ over large patches (sphere/cylinder/plane).
- For conductors in electrostatics, set $E_{\text{inside}} = 0$ and use boundary conditions for surface charge; for dielectrics, prefer symmetry and superposition without advanced $\vec{D}$ formalism.
- Choosing a Gaussian surface that doesn't match symmetry, forcing difficult integrals. Fix: reselect surface to exploit symmetry.

**Example 4.** Using Gauss's law, find $E(r)$ outside a uniformly charged sphere of radius $R$ and total charge $Q$.

**Solution**

Gaussian sphere: $E_r \cdot 4\pi r^2 = Q/\varepsilon_0$, so $\vec{E}(r) = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{\mathbf{r}}$ for $r \geq R$. The magnitude uses $|Q|$.

**Related problems:** No selected problems in this unit yet.

# 9 DC Circuits

*Coverage:* Ohm's law and power; series/parallel reductions; Kirchhoff (KCL/KVL); RC qualitative; RC exact

## 9.1 Ohm's Law and Reductions Core

**Concepts and Formulas**

**Concept explanation:** Ohm's law relates voltage, current, and resistance; power forms help rank dissipation; series/parallel laws simplify networks.
**Core formulas:**

$$\begin{cases} V = IR, \quad P = IV = I^2R = \dfrac{V^2}{R}, \\ R_s = \sum R_i, \quad \dfrac{1}{R_p} = \sum \dfrac{1}{R_i}. \end{cases}$$

**Variable definitions:** $V$ voltage; $I$ current; $R$ resistance; $P$ power.
**Prerequisites & scope:** Ohmic elements only; temperature dependence ignored unless specified.

**Methods and Pitfalls**

- Reduce networks with series/parallel and symmetry; then solve KCL/KVL for the rest of unknowns alone.
- Equate power by using $I^2R$ or $V^2/R$ based on conditions of fixed current/voltage.
- In bridge-type circuits, look for equal potentials in a branch (through symmetry or KCL) to remove it.
- Mixing fixed-voltage and fixed-current contexts when comparing brightness. Fix: pick the power form consistent with constraints.

**Example 1.** Three resistors of $R$ are in parallel; find the equivalent resistance.

**Solution**

$\dfrac{1}{R_{eq}} = \dfrac{1}{R} + \dfrac{1}{R} + \dfrac{1}{R}$ so $R_{eq} = R/3$.

## 9.2 Kirchhoff Laws (KCL/KVL) Core

### Concepts and Formulas

**Concept explanation:** Kirchhoff's current law expresses charge conservation at a node; Kirchhoff's voltage law tracks potential changes around a loop.
**Core formulas:**

$$\begin{cases} \text{KCL: } \sum I_{\text{in}} = \sum I_{\text{out}} \text{ (at a node)}, \\ \text{KVL: } \sum \Delta V = 0 \text{ (around a loop)}. \end{cases}$$

**Prerequisites & scope:** The simple loop-voltage rule assumes negligible changing magnetic flux through the loop; choose current directions and voltage signs consistently.

### Methods and Pitfalls

- Label nodes and choose current directions before writing equations.
- Reduce simple series and parallel branches before solving the remaining equations.
- Mixing voltage rises and drops around a loop. Fix: traverse the loop in one direction and use a consistent sign convention.

**Example 2.** A 12 V ideal battery drives two resistors of 2 Ω and 4 Ω in series. Find the current and both voltage drops.

**Solution**

KVL gives $12 - I(2+4) = 0$, so $I = 2\,\text{A}$. The voltage drops are 4 V and 8 V, whose sum is 12 V.

## 9.3 RC Transients (First Order) Core

### Concepts and Formulas

**Core results:**

$$\begin{cases} \tau = RC, \\ \text{Charge to } V: \ V_C(t) = V + \big(V_0 - V\big)e^{-t/RC}, \\ \text{Discharge to 0: } V_C(t) = V_0 e^{-t/RC}, \\ \text{General: } V_C(t) = V_\infty + (V_0 - V_\infty)e^{-t/\tau}. \end{cases}$$

For these basic circuits $\tau = RC$; in a general linear first-order network use the resistance seen by the capacitor with independent sources suppressed. Capacitor voltage is continuous at switching.

**Example 3.** An initially uncharged capacitor in a series $RC$ circuit has $V$ applied at $t = 0$, with $R = 2\,\Omega$ and $C = 1\,\mathrm{F}$. Find $V_C(t)$.

**Solution**

$V_C(t) = V(1 - e^{-t/RC}) = V(1 - e^{-t/(2\,\mathrm{s})})$.

**Example 4.** A capacitor initially charged to $V_0$ discharges through $R = 2\,\Omega$ with $C = 1\,\mathrm{F}$. Find $V_C(t)$.

**Solution**

The time constant is $RC = 2\,\mathrm{s}$. Thus $V_C(t) = V_0 e^{-t/(2\,\mathrm{s})}$, which starts at $V_0$ and tends to zero.

**Related problems:** Problem 7 (p. 6), Problem 9 (p. 8), Problem 12 (p. 10), Problem 22 (p. 17)

# 10 Magnetism & Induction

*Coverage:* Lorentz force (charges/wires); Faraday-Lenz induction; EM spectrum; Ampere law

## 10.1 Lorentz Force [Core]

**Concepts and Formulas**

**Concept explanation:** A moving charge feels $q\vec{E}$ and $q\,\vec{v} \times \vec{B}$; the magnetic force stays perpendicular to velocity, so it deflects direction without changing speed.
**Core formulas:**

$$\left\{ \vec{F} = q\,(\vec{E} + \vec{v} \times \vec{B}) \right.$$

**Variable definitions:** $q$ charge; $\vec{v}$ particle velocity; $\vec{E}, \vec{B}$ fields.
**Prerequisites & scope:** Nonrelativistic; right-hand rule for cross products.

**Methods and Pitfalls**

- Use right-hand rule consistently; reverse direction for negative charges.
- Magnetic force does no work (always perpendicular to $\vec{v}$), so magnetic fields alone cannot change particle speed.
- Using $q$'s sign incorrectly in $q\,\vec{v} \times \vec{B}$. Fix: compute direction for positive charge, then flip if $q < 0$.

**Example 1.** A particle of charge $q$ enters a uniform magnetic field $\vec{B}$ perpendicular to its velocity with speed $v$. Find the radius and period of its circular motion (neglect $\vec{E}$).

**Solution**

Magnetic force provides the centripetal force: $|q|vB = \dfrac{mv^2}{r} \Rightarrow r = \dfrac{mv}{|q|B}$. The period is $T = \dfrac{2\pi m}{|q|B}$; the sign of $q$ determines the direction of rotation.

## 10.2 Magnetic Force on Wires [Core]

**Concepts and Formulas**

A straight current-carrying wire in a uniform magnetic field experiences a force perpendicular to both current and field. **Core formulas:**

$$\vec{F} = I\,\vec{L} \times \vec{B}, \quad |\vec{F}| = ILB\sin\theta.$$

**Variable definitions:** $I$ current; $\vec{L}$ directed along the current segment with magnitude $L$; $\vec{B}$ magnetic field; $\theta$ angle between $\vec{L}$ and $\vec{B}$.

**Methods and Pitfalls**

- For loops, integrate $d\vec{F} = I\,d\vec{l} \times \vec{B}$ and exploit symmetry.

**Example 2.** A wire of length $L$ carries current $I$ in a uniform field $\vec{B}$ perpendicular to the wire. Find the magnitude of magnetic force.

**Solution**

$F = ILB$.

## 10.3 Faraday-Lenz [Core]

### Concepts and Formulas

**Concept explanation:** Changing magnetic flux induces an emf that opposes the change (Lenz); steady currents set magnetic fields constrained by Ampère's law.
**Core formulas:**

$$\begin{cases} \mathcal{E} = \oint_{\partial S} \vec{E} \cdot d\vec{l} = -\dfrac{d\Phi_B}{dt}, \quad \Phi_B = \iint_S \vec{B} \cdot d\vec{A}, \quad \Phi_E = \iint_S \vec{E} \cdot d\vec{A}, \\ \text{(see Ampère/Maxwell–Ampère for magnetic circulation)} \end{cases}$$

**Variable definitions:** $\mathcal{E}$ induced emf (scalar, $\mathcal{E} = \oint_{\partial S} \vec{E} \cdot d\vec{l}$); $\Phi_B = \iint_S \vec{B} \cdot d\vec{A}$ magnetic flux; $\Phi_E = \iint_S \vec{E} \cdot d\vec{A}$ electric flux; $I_{enc}$ enclosed current.
**Prerequisites & scope:** The displayed electric-field circulation applies to a stationary loop; a moving loop also requires the motional contribution. For time-varying fields use Maxwell–Ampère with displacement current.

### Key Insight

**Terminology note**: *Electric circulation* refers to the line integral of the electric field that defines emf, $\mathcal{E} = \oint_{\partial S} \vec{E} \cdot d\vec{l}$. *Magnetic circulation* refers to $\oint \vec{B} \cdot d\vec{l}$ as used in Ampère/Maxwell–Ampère. These are distinct: $\mathcal{E}$ is a scalar (emf), while $\vec{E}$ and $\vec{B}$ are fields.

### Methods and Pitfalls

- Sketch the loop and determine the positive normal; apply Lenz's rule to deduce the direction of the induced current.
- Apply circular/rectangular Amperian loops along symmetry for infinite wires/solenoids.
- Missing displacement current for charging capacitors. Fix: include $\varepsilon_0 \, d\Phi_E/dt$ in Maxwell–Ampère when fields vary.

**Example 3.** Through a stationary loop of area $A$, a uniform perpendicular magnetic field changes as $B(t) = B_0 + kt$. Find the magnitude of the induced emf.

### Solution

$$|\mathcal{E}| = \left|\frac{d\Phi}{dt}\right| = \left|\frac{d(BA)}{dt}\right| = |k|A.$$

## 10.4 Ampere and Maxwell–Ampere Extension

### Concepts and Formulas

Magnetic circulation is related to enclosed current and, for time-varying fields, to electric-flux change. **Core formulas:**

$$\begin{cases} \oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} \text{ (steady currents)}, \\ \text{Maxwell–Ampere (general): } \oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} + \mu_0 \varepsilon_0 \dfrac{d\Phi_E}{dt}. \end{cases}$$

## 10.5 EM Spectrum and Maxwell (Concept) Core

### Concepts and Formulas

**Concept explanation:** Electromagnetic waves range from radio to gamma; Maxwell's equations couple $\vec{E}$ and $\vec{B}$ and give wave speed $c$ in vacuum.
**Formulas & Concepts:**

$$\begin{cases} c = \dfrac{1}{\sqrt{\mu_0 \varepsilon_0}}, \\ \text{Spectrum ordering by frequency: radio} \rightarrow \text{microwave} \rightarrow \text{IR} \rightarrow \text{visible}, \\ \text{then UV} \rightarrow \text{X} \rightarrow \text{gamma}. \end{cases}$$

**Prerequisites & scope:** Vacuum relations shown; material dispersion alters speed and wavelength.

### Methods and Pitfalls

- Recall typical sources: antennas (radio), thermal radiation (IR), electronic transitions (visible/UV), inner-shell transitions (X-rays), and nuclear transitions (gamma rays).
- In a medium use $v = f\lambda$ and $v = c/n$; $c = f\lambda$ applies to the vacuum wavelength.

**Example 4.** Light in vacuum has wavelength $\lambda = 600\,\text{nm}$ and speed $c = 3 \times 10^8\,\text{m/s}$. Find its frequency.

### Solution

$$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{600 \times 10^{-9}} = 5 \times 10^{14}\,\text{Hz}.$$

**Example 5.** Rank the following by increasing photon energy: radio, visible, X-ray.

**Solution**

Higher frequency means higher photon energy $E = hf$. Ordering: radio < visible < X-ray.

**Related problems:** Problem 8 (p. 7)

# Part IV: Optics & Modern Physics

# 11 Optics

*Coverage:* Reflection; refraction (Snell); thin lens and magnification; interference/diffraction

## 11.1 Reflection and Refraction Core

**Concepts and Formulas**

**Concept explanation:** Light reflects with equal incident and reflected angles; refraction across media obeys Snell's law.
**Core formulas:**

$$\begin{cases} \theta_i = \theta_r, \\ n_1 \sin\theta_1 = n_2 \sin\theta_2, \quad \text{TIR when } \theta_1 > \theta_c = \arcsin(n_2/n_1)\ (n_1 > n_2). \end{cases}$$

**Variable definitions:** $n$ refractive index; $\theta$ angles measured to the normal.
**Prerequisites & scope:** Geometric optics regime; isotropic media; polarization effects ignored here.

**Methods and Pitfalls**

- Draw the normal and principal rays first; search for total internal reflection when going to a lower-$n$ medium.
- Use reversibility of light to validate constructions.

**Example 1.** Light travels from air ($n_1 = 1$) into water ($n_2 = 1.33$) at incidence angle $\theta_1 = 40°$. Find the refraction angle $\theta_2$.

**Solution**

Snell: $n_1 \sin\theta_1 = n_2 \sin\theta_2 \Rightarrow \sin\theta_2 = \dfrac{\sin 40°}{1.33} \Rightarrow \theta_2 \approx 28.9°$.

**Example 2.** Light moves from glass ($n = 1.5$) to air ($n = 1$). Find the critical angle for total internal reflection.

**Solution**

$$\theta_c = \arcsin \frac{n_2}{n_1} = \arcsin \frac{1}{1.5} \approx 41.8°.$$

## 11.2 Thin Lenses and Sign Conventions [Core]

**Concepts and Formulas**

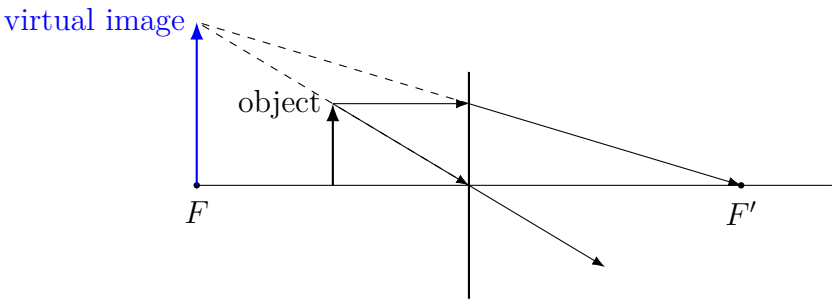
**Concept explanation:** Thin lens imaging follows the lens equation with sign conventions; magnification uses image and object sizes/orientations.
**Core formulas:**

$$\begin{cases} \dfrac{1}{f} = \dfrac{1}{s} + \dfrac{1}{s'}, \quad m = -\dfrac{s'}{s} = \dfrac{h'}{h}. \end{cases}$$

**Variable definitions:** $f$ focal length; $s$ object distance; $s'$ image distance; $m$ magnification; $h', h$ image/object heights.
**Prerequisites & scope:** Use consistent sign convention (e.g., real is positive); paraxial approximation.

**Methods and Pitfalls**

- Combine equation + ray diagram: draw two principal rays to confirm the algebraic image location.
- Remember that negative $m$ indicates inversion; $|m| > 1$ indicates magnification.
- Sign convention (real-is-positive): take $s > 0$ for real objects and $s' > 0$ for real images on the opposite side of the lens from the object; $s' < 0$ indicates a virtual image on the object side (then $m > 0$ and the image is upright).



**Example 3.** An object at $s = 30\,\text{cm}$ forms an image at $s' = -60\,\text{cm}$ using a thin lens. Find the focal length $f$ and magnification $m$.

**Solution**

Lens equation: $\frac{1}{f} = \frac{1}{s} + \frac{1}{s'} = \frac{1}{30} + \frac{1}{-60} = \frac{1}{60}$, so $f = 60\,\text{cm}$. Magnification: $m = -s'/s = -(-60)/30 = 2$. Since $f > 0$, it is a converging lens. Since $s' < 0$, the image is virtual. Since $m > 0$, the image is upright.

## 11.3 Interference and Diffraction Core

**Concepts and Formulas**

**Concept explanation:** Coherent sources produce interference patterns; finite apertures make the light diffract, establishing angular scales by wavelength/aperture.
**Core formulas:**

$$\begin{cases} \text{Double-slit maxima: } d\sin\theta = m\lambda, \\ \text{Single-slit minima: } a\sin\theta = m\lambda,\ m = \pm1, \pm2, \ldots \end{cases}$$

**Variable definitions:** $d$ slit separation; $a$ slit width; $\lambda$ wavelength; $\theta$ diffraction angle.
**Prerequisites & scope:** Small-angle approximations $\sin\theta \approx \theta$ valid near axis; coherence required for stable fringes.

**Methods and Pitfalls**

- Map angles to screen positions with $y \approx L\tan\theta \approx L\theta$ for small $\theta$.
- To resolve features, compare $\lambda$ to $a$ and $d$ to predict fringe spacing/envelope width.

**Example 4.** For double-slit with spacing $d$ and wavelength $\lambda$, what is the angle of the $m$-th bright fringe?

**Solution**

$d\sin\theta = m\lambda \Rightarrow \theta = \arcsin(m\lambda/d)$ (small-angle: $\theta \approx m\lambda/d$).

**Related problems:** Problem 21 (p. 16), Problem 25 (p. 19), Problem 34 (p. 26)

# 12 Modern Physics

*Coverage:* Special relativity ($\gamma$, time dilation, length contraction, $E = mc^2$); photoelectric effect; atomic spectra; nuclear decay/half-life

## 12.1 Special Relativity [Core]

### Concepts and Formulas

**Concept explanation:** At high speeds, time dilates and lengths contract; energy–mass equivalence relates rest mass to rest energy.
**Core formulas (with proper vs observed):**

$$\begin{cases} \gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}, \\ \text{Time dilation: } \Delta t = \gamma\, \Delta\tau \quad (\Delta\tau \text{ proper time in moving clock's frame}), \\ \text{Length contraction: } L = \dfrac{L_0}{\gamma} \quad (L_0 \text{ proper length measured at rest with the rod}), \\ \text{Relativistic energy: } E = \gamma mc^2 \ (E_0 = mc^2), \quad p = \gamma m v, \quad E^2 = (pc)^2 + (mc^2)^2. \end{cases}$$

**Variable definitions:** $\gamma$ Lorentz factor; $v$ relative speed; $c$ speed of light; $\Delta\tau$ proper time (clock's rest frame); $L_0$ proper length (object's rest frame); $E$ total energy; $E_0$ rest energy; $p$ relativistic momentum.
**Prerequisites & scope:** Inertial frames; $v$ along one axis for simple forms; proper quantities measured in an object's rest frame.

### Methods and Pitfalls

- Label frames (S, S') and identify proper time/length before applying formulas.

[Extension] Approximation for $v \ll c$: $\gamma \approx 1 + \frac{1}{2}(v/c)^2$ (derivation and series methods belong to [Extension]).

**Example 1.** A spaceship moves at $0.8c$ relative to Earth. What factor relates proper time to dilated time?

**Solution**

$\gamma = 1/\sqrt{1 - 0.8^2} = \frac{5}{3}$.

## 12.2 Photoelectric Effect Core

### Concepts and Formulas

**Concept explanation:** Electrons emit when photon energy exceeds the work function; the threshold frequency is $f_{th} = \phi/h$.
**Core formulas:**

$$\begin{cases} K_{\max} = hf - \phi, & f_{\text{th}} = \phi/h, \end{cases}$$

**Variable definitions:** $h$ Planck constant; $\phi$ work function.
**Prerequisites & scope:** Idealized models; surface effects and detector thresholds may alter observed $K_{\max}$.

### Methods and Pitfalls

- Stopping potential depends on frequency (threshold via $f_{\text{th}} = \phi/h$), not intensity.
- In a $K_{\max}$–$f$ plot, slope $= h$, vertical intercept $= -\phi$.
- Increasing intensity raises saturation current but does not change stopping potential.

**Example 2.** Light of frequency $f$ hits a metal with work function $\phi$. Write the maximum kinetic energy of ejected electrons.

### Solution

If $hf \geq \phi$, then $K_{\max} = hf - \phi$. If $hf < \phi$, no photoelectrons are emitted in the ideal model.

## 12.3 Nuclear Decay Basics Core

### Concepts and Formulas

For a constant decay rate, the number of undecayed nuclei decreases exponentially:
**Core formulas:**

$$N(t) = N_0\, 2^{-t/T_{1/2}} = N_0 e^{-\lambda t},\ \lambda = \frac{\ln 2}{T_{1/2}}.$$

$N_0$ is the initial number, $T_{1/2}$ is the half-life, and $\lambda$ is the decay constant.

### Methods and Pitfalls

- For decay chains, use activity $A = \lambda N$; independent branches superpose exponentials.
- Plot $\ln N$ vs $t$ to extract $\lambda$ from the slope.

**Related problems:** Problem 32 (p. 25)